

HOW THANKSGIVING AND SUPER BOWL TRAFFIC CONTRIBUTE TO FLIGHT
DELAYS

by

XXXXX

A Graduate Capstone Project Submitted to the College of Aeronautics,
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This Graduate Capstone Project was prepared under the direction of the candidate's Graduate Capstone Project Chair, XXXXX, Worldwide Campus, and has been approved. It was submitted to the Department of Graduate Studies in partial fulfillment of the requirements for the degree of Master of Science in Aeronautics

Graduate Capstone Project:

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Graduate Capstone Project Chair

Date

Acknowledgements

I'd like to thank my legs, for always supporting me; my arms, who are always by my side; and lastly my fingers, I can always count on them.

Abstract

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Title: How Thanksgiving and Super Bowl Traffic Contribute to Flight Delays

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This study explores the effects of non-scheduled flights on scheduled flight delays during Thanksgiving and Super Bowl across 5 years. Flight delay data were collected from the Bureau of Transport Statistics and the Federal Aviation Administration. Super Bowl and Thanksgiving were chosen as the special events of interest for this study as they provided complementary datasets. Super Bowl showed an increase in non-scheduled flights whereas Thanksgiving showed greater scheduled flight operations. The results of this study concluded that scheduled flights showed greater delays during Super Bowl when compared to Thanksgiving. A significant interaction was also found to exist between scheduled and non-scheduled flights operating during the two special events. Both scheduled flight delays and non-scheduled flight delays increased during Super Bowl. However, during Thanksgiving this relationship did not exist – scheduled flights had much fewer delays than non-scheduled flights. Due to the increase in the number of non-scheduled flight operations during Super Bowl, delays increased thereby increasing operating costs for flights. The outcomes of this study shed light on another aspect of airspace efficiency that could be researched to reduce costs and improve user experience. To mitigate these potential issues, it is important all pilots receive special training to predict potential bottlenecks and delays while operating at airports hosting special events.

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Chapter I

Introduction

Flight delays associated with special events during the year require additional planning by Air Traffic Control (ATC). The National Airspace System (NAS) implements Special Traffic Management Programs (STMPs) for special events such as holiday travel during Thanksgiving and the Super Bowl (Krozel, Hoffman, Penny, & Butler, 2003). Additional flights scheduled by airlines during these special events saturate the airspace and the airports in the vicinity of where the special event is taking place. Congested airspace may lead to greater delays for the traveling public. Specialists in the Air Traffic Control System Command Center (ATCSCC) note that travel on the Sunday following Thanksgiving increases significantly (Krozel et al., 2003). Similarly, traffic increases following Super Bowl Sunday. Therefore, exploring the interaction of delays associated with specific types of flights operating during special events may aid in optimizing the throughput of the NAS.

This study focused on scheduled airline flights and how they are affected during special events. Factors that were considered in this study included non-scheduled flights such as private or charter flights. Clare and Richards (2013) state that traffic flow management relies on predictions of demand in the airspace system. This can become difficult to manage due to factors such as unscheduled demand at these special events which may vary yearly. Inefficient traffic flow may cause greater delays and costs for scheduled air carrier service which propagate to operations across the entire NAS. The Super Bowl and Thanksgiving weekend were used to analyze the effects of non-scheduled flights on scheduled operations across multiple years.

Significance of the Study

The significance of this study was to discover the effects on scheduled flights during special events. This research can assist the ATC system to manage, direct, and mitigate potential delays and congestion associated with special events during the year. The results of this study may also be able to assist airlines in developing flight schedules that consider the effects of non-scheduled flights. Using this information, airlines and the Federal Aviation Administration (FAA) can decrease costs by reducing delays. By exploring factors, such as types of flights, stakeholders in aviation can make changes to mitigate or even eliminate the negative impacts of flight delays during special events.

Statement of the Problem

Flight delays have become a prominent issue in the aviation industry as it grows in traffic and density. A steady growth has been occurring in the number of aircraft that are flying in the NAS. With a thorough understanding of why delays occur, appropriate modifications to the airspace system can be made to increase capacity. There are many studies that look at flight delays in the NAS, however, not many evaluate delays associated with special events and effects of non-scheduled flights. Using delay data from the Department of Transportation (DOT) and the FAA, the researcher will better understand delays associated with scheduled and non-scheduled flights.

Purpose Statement

The aim of this study was to compare flight delays of scheduled flights during Thanksgiving versus Super Bowl, when there is a greater rate of non-scheduled flights. By comparing delays during these two events, a greater understanding of the effects of non-scheduled flights on scheduled flights can be achieved.

Research Question

Using Airline Delay Data from the Bureau of Transport Statistics (BTS) and Non-Scheduled Flight Delays from the FAA, statistical tests were conducted to analyze if a significant increase in delays occurs during special events. The question posed in this study is to discover if delays exist for scheduled flights due to an increase in non-scheduled flights during Super Bowl compared to Thanksgiving.

Delimitations

The study was limited to observing Thanksgiving weekend and Super Bowl weekend. These special events were singled out as they provided two complementary datasets. Thanksgiving provides a similar number of total flights operating in the NAS compared to Super Bowl, while, the Super Bowl provides a greater number of non-scheduled flights compared to Thanksgiving.

Limitations and Assumptions

The airports used in the study were limited to locations in the vicinity of the following Super Bowl. Total flight time for non-scheduled flights was assumed to be from doors closed to doors open. Therefore, delay data was calculated based on the total flight time specified from the ATC system. Flights were analyzed from 2010 to 2015 due to the limitations in acquiring non-scheduled data from the FAA.

List of Acronyms

ATC	Air Traffic Control
ATCSCC	Air Traffic Control System Command Center
BTS	Bureau of Transport Statistics
DOT	Department of Transportation

FAA	Federal Aviation Administration
GDP	Gross Domestic Product
MIT	Miles in Trail
NAS	National Airspace System
NBAA	National Business Aviation Association
STMP	Special Traffic Management Program
TFMS	Traffic Flow Management System

Chapter II

Review of the Relevant Literature

The definition of a delayed flight according to the FAA (2015) is the departure or arrival time exceeded by 15 minutes when compared to the scheduled departure or arrival time. These delays cause expenses to rise for airlines operating with very thin margins. Additionally, passengers must bear the cost of delays as well in the form of time lost in productivity. A review of research conducted about the causes of flight delays and the associated economic impacts will be reviewed in this chapter. Furthermore, the impacts of non-scheduled flights on the NAS will be examined to better understand airspace congestion.

Reasons for Delay

Several factors can cause delays to increase in local environments expanding to the entire country. These delays can be caused by a variety of reasons ranging from late arriving aircraft to passenger or baggage delays. With the growing numbers of air traffic, airspace congestion is becoming a larger issue to solve. The FAA (2009) forecast a 50% increase in flight operations between 2010 and 2025. Unless new procedures are put in place, the amount of delay encountered by aircraft will only become worse as the traffic density increases.

Congestion in air traffic. Most delays are encountered around bottlenecks in the airspace system. These usually occur due to limited number of runways available for aircraft to utilize. Glockner (1996) states that one method of reducing congestion would be to increase landing capacity. Flights operating to airports in cities hosting special events would experience these delays. Events such as the Super Bowl, National Business

Aviation Association (NBAA) conferences, Oshkosh Fly-In, etc. can cause excessive delays due to congestion.

Flow management tries to alleviate some of the delays encountered due to overcrowding of terminal airspace or airports. When aircraft reach a point of congestion, ATC starts managing and routing them into a long airborne queue to sequence them into the airport. Rerouting these aircraft or placing them in holding patterns increases the amount of fuel burned for these aircraft, thereby increasing the costs (Glockner, 1996). Additionally, speed reductions and altitude changes required by aircraft en route to a congested airspace may increase fuel consumption and directly incur greater expenses in the current system (Delgado & Pratz, 2014). Therefore, understanding and recognizing potential congestion during certain seasons or events can aid in reducing costs and maintaining efficiency.

Another aspect of congestion stems from the limitations of airport capacity at arrival and departure fixes. These fixes act as gates for entry or exit points for the terminal environment of the airport. Each airport has a limited number of fixes that can be utilized in different runway configurations, resulting in limited capacity for the airport to handle mass arrivals and departures. Gilbo (1997) concludes effective utilization of airport capacity is affected by under and overloaded fixes in the terminal area. Aircraft arriving from one direction in large quantities can cause an airport or a cluster of airports to experience significant delays due to sector saturation.

Extreme Weather. A major expense associated with operating an airline is the cost of flight delays (McCrea, Sherali, & Trani, 2008). Global weather patterns are constantly changing and can be difficult to predict. Hurricanes can grow in strength over

a short period, making it difficult to compensate with the flight schedule. Most extreme weather events require large in flight reroutes or significant delay on the ground to maintain a safe following distance in the air. To achieve this, ATC issues ground holds for aircraft at the airport until sufficient distance is created, allowing for greater spacing in the airspace system. A loss of capacity due to extreme weather causes 93% of flight delays at hub airports according to (Clarke, 1998).

These occurrences of severe weather conditions not only affect the timely arrival or departures of aircraft, but also the effectiveness of air traffic management systems controlling the safe transit of passengers (Walker, Chakrapani, & Elmahboub, 2008). Due to the unpredictability of weather, delays associated with it are harder to mitigate and plan for in advance. The formation of thunderstorms as meteorological events and the impact on aviation safety is another aspect that is of interest in further research for reducing delays.

Growth Resulting in Increased Delays

Aviation is a growing industry requiring greater capacities at airports and in the NAS. Evans & Schäfer (2011) state that the current capacity will be exceeded in the coming years, even with updates planned for the NAS. As the total capacity of the airspace environment reaches close to complete saturation, delays will increase resulting in greater costs for users. Airspace will become more restrictive due to the increase in the number of aircraft operating in the same amount of space. With nearly 1/4 of all flights experiencing delays greater than 15 minutes in 2007, system saturation is slowly becoming an apparent issue that needs to be dealt with soon (Peterson et al., 2013). These

delays will continue to increase until new technology is implemented to increase NAS capacity.

Cost of Flight Delay

The dawn of the modern age of jet aircraft has resulted in airlines accounting for a large part of the Gross Domestic Product (GDP). According to the FAA (2015), the civil aviation industry contributed up to 5.4% of the United States' GDP. Due to the aviation sector having such large impacts on the economy of the country, greater efficiency would be an indispensable factor for profitable airline operations. Flight delays not only affect passengers traveling for business or leisure, but also freight transportation around the country. Both cargo and passengers contribute to the effect aviation as a form of transportation has on the GDP of the country. Schumer (2008), found that the total cost of air traffic delays in the United States cost the economy \$41 billion during the year 2007. The net wealth for the United States would increase by \$17.6 billion with a 10% reduction in flight delay and by \$38.5 billion with a 30% reduction (Peterson et al., 2013). This economic cost includes the direct cost savings for airlines as well as the indirect effects of saved time for passengers and cargo.

Cost to airlines. Airlines in the United States have been able to operate efficiently and effectively to keep operating costs down. From 1995 to 2014, the average fare for an airline itinerary, adjusted for inflation, has increased 34.1% from \$292 to \$392 according to BTS (2015). Amidst such tight cost control and attention to efficiency in order to keep airline ticket prices low, flight delays costs airlines a large percentage of their operating costs. Schumer (2008) states that \$19.1 billion of airline operating costs

were attributed to delays. This is a significant cost for the aviation service providers to absorb and continue offering reasonable rates for transportation of people and goods.

Cost to passengers. Delays affect airlines in operating costs both directly and indirectly. A cost is also incurred to the passengers traveling on these flights. Baik, Li, and Chintapudi (2010) showed flight delays in the domestic market cost passengers \$5.2 billion in 2007. These passengers lost valuable time and productivity for their employers, businesses, or leisure activities due to flight delays (Lubbe & Victor, 2012). Costs were most likely higher as these passengers most likely missed connection flights or had cancellations leading to lost pre-paid hotel accommodations, ground travel plans, and vacation time. It is clear that flight delays have effects that transpire farther than NAS and the airlines.

Effects on other industries. Many other industries rely upon the airline industry, such as food service, lodging, retail, ground transportation, and entertainment. When the aircraft is delayed on the tarmac or in the air, passengers cannot spend that lost time in businesses providing commerce. Because of delays, these industries combined lost about \$10 billion (Schumer, 2008). This number did not include delays caused to cargo being transported by air, which would affect the manufacturing industry more than anything else would. Overall, delays increase costs to all entities involved in air transportation ranging from the airlines to industries reliant upon air transportation for delivering customers and cargo to their destinations.

Flight Delays and Non-Scheduled Flights

Delay management utilizes many procedures and methods to reduce or eliminate delays. Methods such as flow management utilize schedules to maintain efficiency and

on-time performance. Non-scheduled flights that do not publish schedules in advance may cause issues and reduce efficiency of techniques such as flow management. Aircraft travelling in the NAS need spacing separation imposed as Miles-in-Trail (MIT). Ensuring that aircraft pass from one sector of airspace to another with the appropriate spacing for safety causes flights delays (Sheth & Gutierrez-Nolasco, 2014). Research conducted on the effects of non-scheduled flights and delays associated is very limited and further investigation is necessary.

Special Events Traffic

The NAS experiences seasonal, weekly, and daily trends in aircraft movements. Looking at the aggregate statistics, certain trends repeat on a yearly basis. Seasonal trends indicated an increase in air travel during the summer months and during December. This corresponds with holiday travel presenting peaks towards the start of school for students in August. Additionally, Krozel et al. (2003), state a significant impact on the NAS originates from Thanksgiving and Christmas travel. Weekly trends in the NAS display a tri-modal distribution indicating after weekdays, Sunday has the most travel for the week.

These seasonal and weekly trends become a factor when associated with special events such as the Super Bowl or winter travel. Travel during events that occur on Sunday result in an increase in traffic during the Thursday and Friday prior and the Monday after the event (Krozel et al., 2003). These events do not cause a significant strain on the NAS, as they are localized events. However, airports near the Super Bowl or college games experience delays due to airspace congestion. Compare this to increased travel during the summer or winter months, traffic loads increase across the entire NAS. Localized events such as NBAA, Super Bowl, or college football games, etc., have

greater traffic due to on-demand aircraft that are not scheduled. Large scale air traffic increases such as winter or summer holiday travel is predominately scheduled traffic accounted for by the airlines and other scheduled flight operations.

Summary

Many studies analyze the effects of large increases in traffic volume, such as holiday travel, to better model flight schedules and reduce delays. However, there is a lack of research investigating the effects of flight delays for localized events such as the Super Bowl or other events that affect flight delays in metropolitan areas that are severed by a close grouping of airports.

Comparing flight delays between localized events and the entire NAS will provide further insight into certain causes of flight delays not explored before. Results from research evaluating delays for localized events can be used to evaluate current procedures in place and the areas they lack in maintaining efficiency. Furthermore, ATC service providers can use this information to evaluate the circumstances that create the greatest delays and put into place strategies that can actively mitigate these bottlenecks.

Chapter III

Methodology

Research Approach

This study used exploratory and quantitative methods to compute the amount of delays scheduled air carriers experience during special events. *T*-tests were conducted to compare means between special events and types of flights to study the effects on flight delays. Thanksgiving and Super Bowl were the complementary special events that exhibited flight delays associated with scheduled and non-scheduled flights. A greater amount of scheduled traffic operates during the time of Thanksgiving. On the other hand, a shift to more non-scheduled flights is evident during Super Bowl. Analyzing the flight delay experienced during these two special events could benefit operators and service providers with a greater understanding of delays. The flight delay data retrieved from the FAA website and BTS were the primary sources for the analysis.

Design and procedures. This study was determined to be an exploratory and quantitative study. Total flight delay was measured for airlines and non-scheduled flights during two special events through *t*-tests using the On-Time Arrival Performance data from the FAA and BTS. This data contained all scheduled and non-scheduled flights during Thanksgiving and Super Bowl from 2010 to 2015. Super Bowl delays were to be compared with Thanksgiving delays from the previous year to minimize any external influences on traffic. The DOT provides a 15-minute tolerance before a flight is considered delayed. Despite the DOT definition for late flights, data were analyzed based on flights exceeding scheduled total flight time, i.e., a plane scheduled to complete a

flight in 2 hours but actually completing it in 2 hours and 1 minute would be considered 1 minute late.

Flight delay data for airlines was collected and organized from the BTS website. The data was limited to the airports in the vicinity of the Super Bowl venue. Similarly, non-scheduled flight data was collected and organized from the FAA Traffic Flow Management System (TFMS) database for the same airports as the scheduled flights. The data was collected for the day before, the day of, and the day after the Super Bowl and for the Saturday, Sunday, and Monday immediately after Thanksgiving. These days of the week were chosen during Thanksgiving because the Sunday following Thanksgiving is one of the busiest travel days of the year. This approach allowed for consistent traffic between the two special events. Using flight delay data from Thanksgiving allowed for data that provided a greater number of scheduled operations while flight delay data for the Super Bowl provided greater number of non-scheduled operations. Between the two events, each year provided a similar number of total aircraft. *T*-tests were conducted to measure the difference in delays between the events and types of flights.

Apparatus and materials. This project used Microsoft Excel and SPSS to organize, code, and analyze the flight delays. Microsoft Excel was used to organize all the scheduled flights and non-scheduled flights into separate sheets for Thanksgiving and Super Bowl. This data was transferred to SPSS to conduct the descriptive and statistical tests such as *t*-tests and ANOVAs.

Sample

The sample for this study was all air traffic, scheduled and non-scheduled flights, operating to the specific cities during Thanksgiving and Super Bowl from 2010 to 2015.

This data compares total flight delay for scheduled and non-scheduled flights. Data for each special event focused on flights on Saturday, Sunday, and Monday to avoid any unintended inference from traffic fluctuations between the two events and different days of the week for travel. Travel on these 3 days was important because both events have a substantial amount of travel occurring during these times. The Super Bowl occurs on a Sunday and results in an increase in aircraft arrivals and departures around the event. Thanksgiving has historically had the highest amount of travel on the Sunday and Monday following the event. Both events provide complimentary datasets. Most people travel back home after spending time with their families on Thanksgiving and result in a surge of traffic during the weekend and the start of the week. This results in greater scheduled operations by the airlines for this event. Many individuals travel to the Super Bowl the day before or on the day of the game, consequently increasing non-scheduled traffic. Many people are traveling to be back home in time for the start of the week following the event resulting in increased traffic on Monday. This results in greater non-scheduled traffic in the form of charter or private aircraft flying.

Sources of the Data

Data for this study was collected from two different sources. Scheduled airline flight delay data was obtained from the BTS website using the On-Time Performance table. Non-scheduled flight data was acquired from the FAA using their TFMS data that tracks flight delays in the NAS. This data from the FAA was collected by contacting the Air Traffic Control System Command Center (ATCSCC).

Validity

The data collected for this study is assumed to be reliable and valid as it is from a government source. The total flight delay times were calculated from this data, and the reported information is assumed to be reasonable. Scheduled flight times are presumed to be reported appropriately by the airlines for all flights they operate. The non-scheduled flight data was calculated based on the flight plan filed by the crew. The flight times for these non-scheduled aircraft are assumed to be filed with the ATC system accurately and valid for this study. Based on the flight time data from these flight plans, total flight delays were calculated for non-scheduled aircraft.

Treatment of the Data

The raw data gathered from BTS and the FAA TFMS system was organized in excel to compare consistently. The only common factor between both sets of data was total flight time. This was the factor that would be focused on for both events and types of flights. Total flight delay for both sets of data was generated by finding the difference in time between scheduled and actual. Total flight delay was calculated only for scheduled flights which were not cancelled or early.

Descriptive statistics were computed for scheduled and non-scheduled flights. Additional data available to be computed for the scheduled aircraft included departure delay and arrival delay. This data was computed by subtracting the actual time of departure or arrival from the scheduled times. Flights that were cancelled or arrived early were coded as zeroes.

The data was imported into SPSS after all the data was calculated, organized, and coded. Descriptive statistics and *t*-tests were run to determine total number of flights for

type of traffic and event. The average delay was computed for each year between Super Bowl and Thanksgiving as well as types of flights. Once all these statistical tests were run, an ANOVA was run to find any significant interaction between the types of flights and between Super Bowl and Thanksgiving.

Chapter IV

Results

Descriptive Statistics

The dependent variable for this study included total delay, departure delay, taxi out time, and taxi in time. Table 1 shows the descriptive statistics for delays occurring during Super Bowl and Thanksgiving for scheduled and non-scheduled flights.

Table 1

Descriptive Statistics for Delays During Type of Operation and Special Event

Special Event	Type of Operation	N	Min	Max	Mean	SD
Super Bowl	Scheduled	11882	0	157.00	5.50	12.85
	Unscheduled	9314	0	1502.00	4.94	36.26
	Total	21196	0	1502.00	5.25	25.89
Thanksgiving	Scheduled	14668	0	93.00	2.05	5.47
	Unscheduled	4499	0	765.78	3.81	26.99
	Total	19167	0	765.78	2.47	13.95
Total	Scheduled	26550	0	157.00	3.59	9.66
	Unscheduled	13813	0	1502.00	4.57	33.53
	Total	40363	0	1502.00	3.93	21.13

Note. Mean, Min, Max, and *SD* are measured in minutes. Min = Minimum, Max = Maximum, *SD* = Standard Deviation. Data from 2010-2015 were used for this analysis.

Scheduled and Non-Scheduled Comparison

The total number of scheduled and non-scheduled flights by years are displayed in Figure 1. The years are divided by Thanksgiving and Super Bowl. Figure 2 displays the average departure delay, taxi out times, taxi in times, and the total delay encountered by scheduled flights. Average scheduled and non-scheduled flight delays are presented in Figure 3.

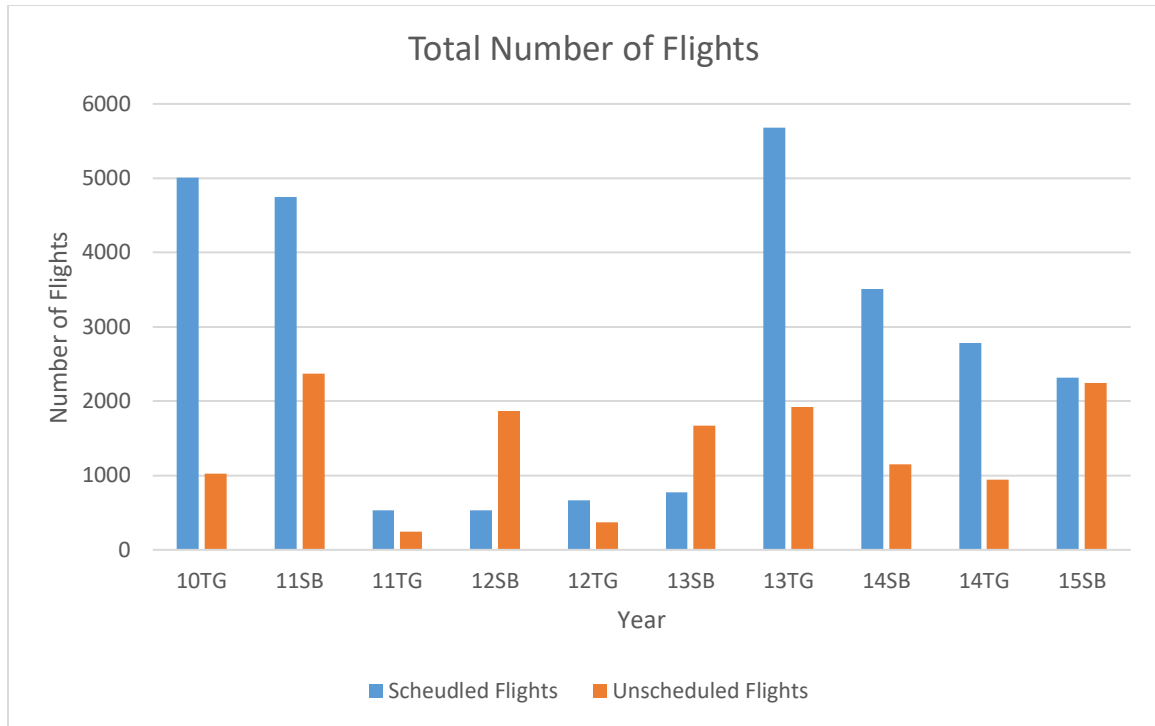


Figure 1. Comparison Between Thanksgiving and Super Bowl for Scheduled and Non-Scheduled flights. TG = Thanksgiving, SB = Super Bowl.

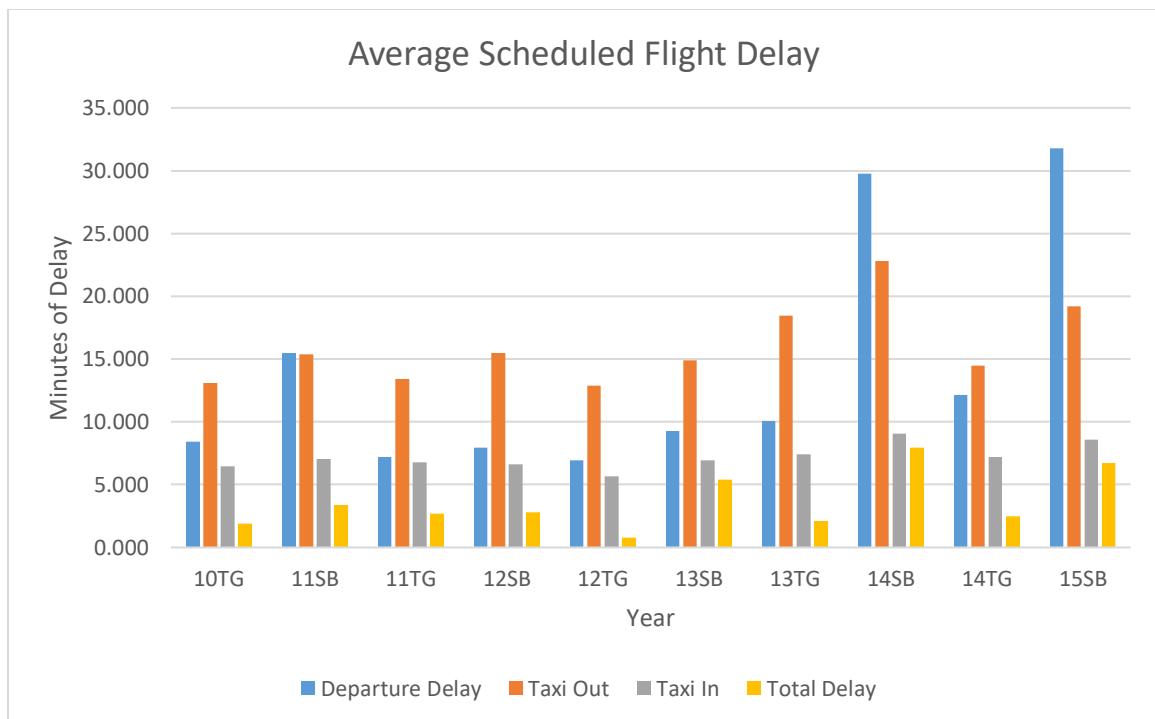


Figure 2. Departure Delays, Taxi Out Times, Taxi In Times, and Total Flight Delays for Scheduled Flights Across Thanksgiving and Super Bowl. TG = Thanksgiving, SB = Super Bowl

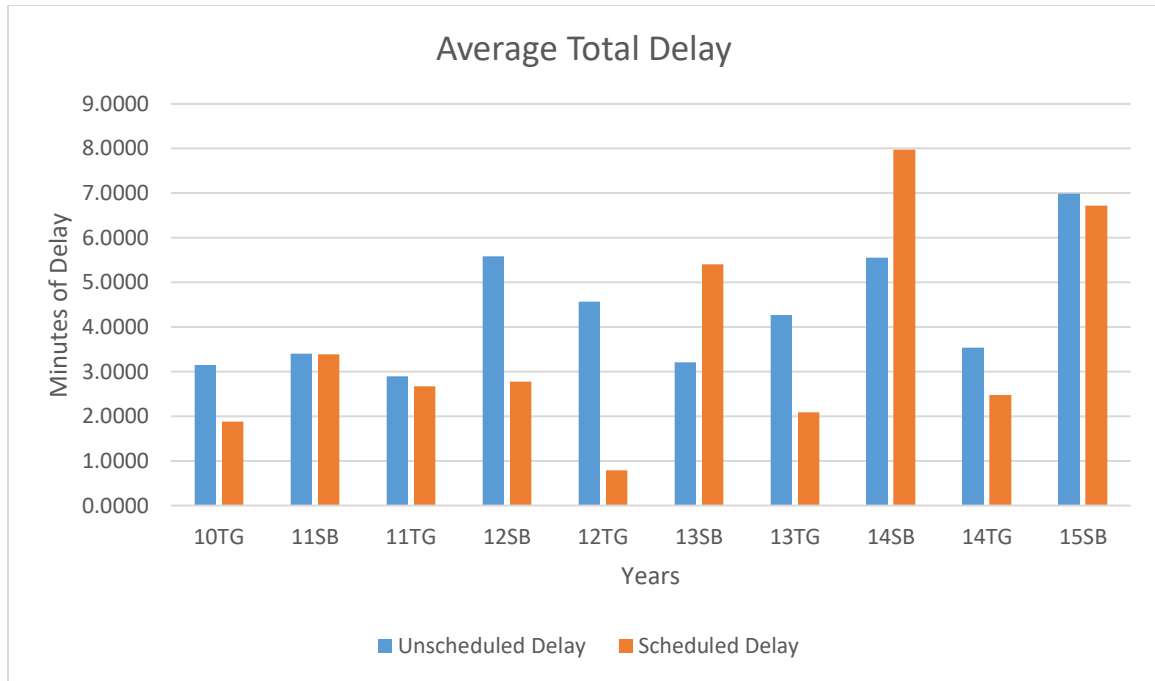


Figure 3. Average Total Delay Compared Between Scheduled and Non-Scheduled Flights. TG = Thanksgiving, SB = Super Bowl.

Super Bowl Delays and Thanksgiving Delays Phoenix (2014 – 2015)

The null hypothesis was that there is no difference in total delay for scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was significant ($p < .05$), and thus, an adjustment to degrees of freedom was made. The mean total delay for Super Bowl ($M = 6.719$, $SD = 12.996$) was longer than the mean total delay for Thanksgiving ($M = 2.473$, $SD = 5.878$). An independent samples t -test was significant, $t(3102.89) = 14.542$, $p < .001$. Therefore, the null hypothesis was rejected.

The null hypothesis was that there is no difference in total delay for non-scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was significant ($p < .05$), and thus, an adjustment to degrees of freedom was made. The mean total delay for Super

Bowl ($M = 6.982$, $SD = 57.384$) was longer than the mean total delay for Thanksgiving ($M = 3.541$, $SD = 22.710$). An independent samples t -test was significant, $t(3180.27) = 2.425$, $p < .05$. Therefore, the null hypothesis was rejected.

Super Bowl Delays and Thanksgiving Delays New York (2013 – 2014)

The null hypothesis was that there is no difference in total delay for scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was significant ($p < .05$), and thus, an adjustment to degrees of freedom was made. The mean total delay for Super Bowl ($M = 7.968$, $SD = 18.007$) was longer than the mean total delay for Thanksgiving ($M = 2.087$, $SD = 5.703$). An independent samples t -test was significant, $t(3944.278) = 18.766$, $p < .001$. Therefore, the null hypothesis was rejected.

The null hypothesis was that there is no difference in total delay for non-scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was significant ($p < .05$), and thus, an adjustment to degrees of freedom was made. The mean total delay for Super Bowl ($M = 5.559$, $SD = 29.981$) was longer than the mean total delay for Thanksgiving ($M = 4.271$, $SD = 29.964$). An independent samples t -test was not significant, $t(2419.688) = 1.152$, $p > .05$. Therefore, the null hypothesis was retained.

Super Bowl Delays and Thanksgiving Delays New Orleans (2012 – 2013)

The null hypothesis was that there is no difference in total delay for scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was significant ($p < .05$), and thus, an adjustment to degrees of freedom was made. The mean total delay for Super Bowl

($M = 5.402$, $SD = 11.052$) was longer than the mean total delay for Thanksgiving ($M = 0.785$, $SD = 2.794$). An independent samples t -test was significant, $t(888.535) = 11.228$, $p < .001$. Therefore, the null hypothesis was rejected.

The null hypothesis was that there is no difference in total delay for non-scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was not significant ($p < .05$) and no adjustments were made to degrees of freedom. The mean total delay for Super Bowl ($M = 3.206$, $SD = 25.948$) was shorter than the mean total delay for Thanksgiving ($M = 4.575$, $SD = 33.034$). An independent samples t -test was not significant, $t(2037) = -0.871$, $p > .05$. Therefore, the null hypothesis was retained.

Super Bowl Delays and Thanksgiving Delays Indianapolis (2011 – 2012)

The null hypothesis was that there is no difference in total delay for scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was not significant ($p < .05$) and no adjustment was made to degrees of freedom. The mean total delay for Super Bowl ($M = 2.769$, $SD = 6.453$) was longer than the mean total delay for Thanksgiving ($M = 2.669$, $SD = 7.039$). An independent samples t -test was not significant, $t(1063) = 0.242$, $p > .05$. Therefore, the null hypothesis was retained.

The null hypothesis was that there is no difference in total delay for non-scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was significant ($p < .05$) and an adjustment was made to degrees of freedom. The mean total delay for Super Bowl ($M = 5.588$, $SD = 25.565$) was shorter than the mean total delay for Thanksgiving

($M = 2.896$, $SD = 28.037$). An independent samples t -test was not significant, $t(293.727) = 1.416$, $p > .05$. Therefore, the null hypothesis was retained.

Super Bowl Delays and Thanksgiving Delays Dallas (2010 – 2011)

The null hypothesis was that there is no difference in total delay for scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was significant ($p < .05$), and thus, an adjustment to degrees of freedom was made. The mean total delay for Super Bowl ($M = 3.392$, $SD = 7.493$) was longer than the mean total delay for Thanksgiving ($M = 1.888$, $SD = 5.002$). An independent samples t -test was significant, $t(8212.155) = 11.597$, $p < .001$. Therefore, the null hypothesis was rejected.

The null hypothesis was that there is no difference in total delay for non-scheduled flights between Super Bowl and Thanksgiving. The assumption of equality of variance was tested. Levene's test of equality of variance was not significant ($p < .05$) and no adjustment was made to degrees of freedom. The mean total delay for Super Bowl ($M = 3.407$, $SD = 24.677$) was longer than the mean total delay for Thanksgiving ($M = 3.143$, $SD = 21.559$). An independent samples t -test was not significant, $t(3398) = 0.297$, $p = > .05$. Therefore, the null hypothesis was retained.

Comparing Special Event and Type of Operation

The objective was to find if there was a difference in delays based on the Type of Operation and Special Event. The null hypothesis was that there was no difference in delay times based on type of operation and special event. Levene's test of equality of variances was significant ($p < .05$), indicating a violation of homogeneity of variance.

Due to a significant Levene's statistic ($p < .05$), the significances of the main effects and interaction should be interpreted with caution.

A 2 (Type of Operation: Scheduled, Non-scheduled) x 2 (Special Event: Super Bowl, Thanksgiving) two-way between-subjects ANOVA was conducted on delay times. The results showed a significant main effect of Type of Operation, $F(1, 40362) = 6.723$, $p < .05$, partial eta-squared $< .001$, a significant main effect for Season, $F(1, 40362) = 97.452$, $p < .05$, partial eta-squared $= .002$, and a significant Type of Operation x Season interaction, $F(1, 40362) = 25.044$, $p < .05$, partial eta-squared $= .001$. The null hypothesis was rejected.

The results of the simple main effects post hoc tests indicated that the mean delays during Super Bowl for all flights ($M = 5.251$, $SD = 25.894$) was significantly higher than the mean for all flights during Thanksgiving ($M = 2.467$, $SD = 13.946$). The post hoc tests of the main effect of Type of Operation indicated the mean for Non-scheduled flights ($M = 4.571$, $SD = 33.531$) was significantly higher than the means for Scheduled flights ($M = 3.594$, $SD = 9.663$). Figure 4 shows a significant interaction between the Type of Operation and Special Event.

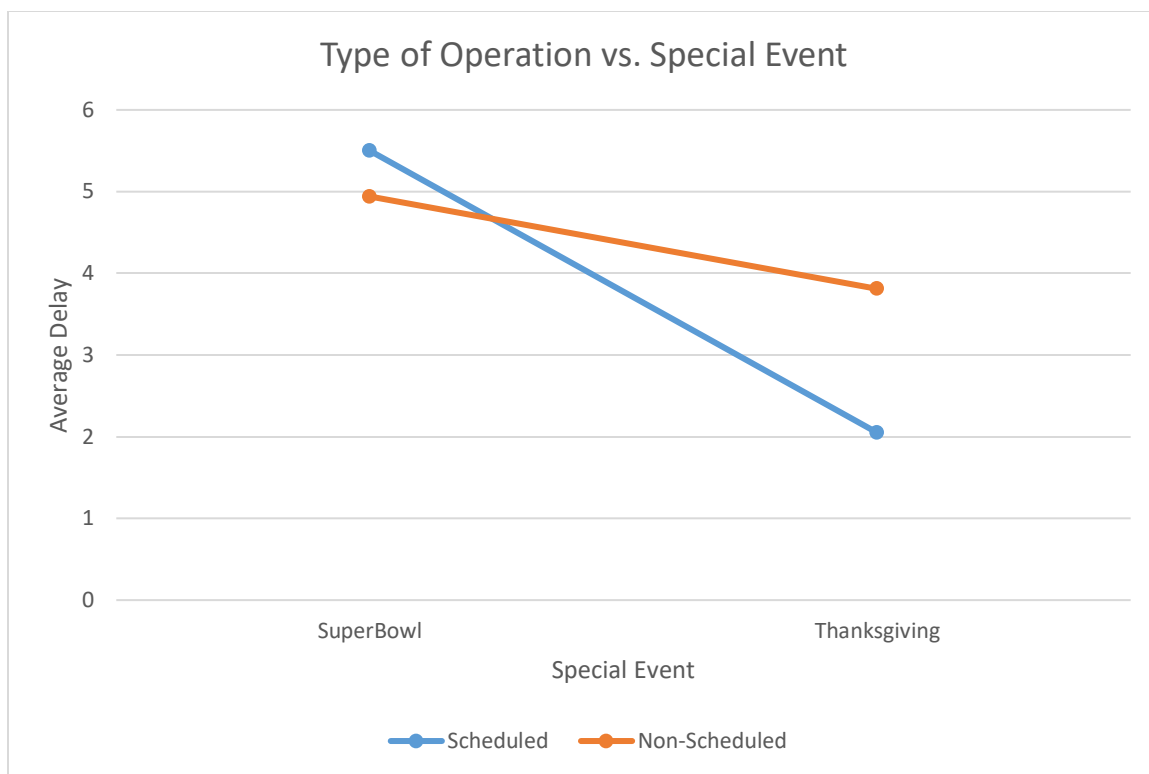


Figure 4. Interaction between Type of Operation and Special Event.

Chapter V

Discussion, Conclusions, and Recommendations

The results from Chapter IV helped the researcher identify certain trends and noteworthy patterns for scheduled flight delays between the Super Bowls and Thanksgivings.

Discussions

The total number of flight for Super Bowl and Thanksgiving were similar. However, Super Bowl had almost twice as many non-scheduled flight as compared to Thanksgiving. About two-thirds of the total flights were scheduled flights while the remaining one-third made up the non-scheduled flights. The lower number of non-scheduled flights during Thanksgiving may have been a causal factor in the lower average delay when compared to during the Super Bowl.

Across all the Super Bowls, non-scheduled flights increased when compared to Thanksgiving. This increase was clear when comparing all years, except when the event took place in New York and was affected by a snowstorm. These years also indicated slight decreases in the total number of scheduled flights during the Super Bowl related to Thanksgiving.

Analyzing only the scheduled flight delays based on departure delay, taxi out and taxi in times, and total delay, an increase in all these times can be observed during Super Bowl over Thanksgiving. Similarly, comparing the total delay between the two special events, scheduled flight delays increased significantly during Super Bowl.

Out of the five venues, only Indianapolis did not have a significant difference between the Super Bowl and Thanksgiving for scheduled traffic. Indianapolis had fewer

number of flights operating at the airport and may be the reason it did not result in a significant difference. Similarly, for non-scheduled traffic, only Phoenix had a significant difference between Super Bowl and Thanksgiving. For the non-scheduled flights at Phoenix, there was a large disparity between Super Bowl and Thanksgiving. This drastic change in the number of flights may have been the cause for significant difference between the amounts of delay experienced.

There exists a significant interaction between the type of operation and the special event. During Thanksgiving scheduled flight had significantly lower delays than non-scheduled flights. However, during Super Bowl, both types of operations yielded no significant difference in delay statistics.

Conclusions

Super Bowl had greater delays overall than Thanksgiving. The delays for Super Bowl increased mostly when there was an increase in non-scheduled flights. One reason may be because non-scheduled flights do not regularly operate to the venues where the Super Bowl was held; therefore, resulting in an increase in traffic, which can cause bottlenecks. These bottlenecks can occur for reasons ranging from saturated airspace to limited parking spaces.

The increase in non-scheduled flights during Super Bowl is a trend that can be observed across the five venues. This is most likely due to fans travelling using private or personal aircraft to the Super Bowl. These fans have the luxury of arriving before the start of the event and departing after the game has ended. The concentrated arrivals and departures may also cause delays due to the demand meeting or even exceeding capacity for the airspace and airport.

Thanksgiving provides slightly different characteristics for non-scheduled travel compared to Super Bowl. The number of non-scheduled flights arriving and departing during the Thanksgiving holiday season is spread out over a greater time period and therefore does not affect delays as drastically. Furthermore, travel is not concentrated at a specific locale for the holiday and does not contribute to delays.

Scheduled flights usually operate efficiently as they are supported by a vast network from the airlines, yielding in fewer delays. Non-scheduled flights, on the other hand, do not have dedicated resources and rely on local facilities that result in slightly greater delays. This trend can be clearly observed in delays during Thanksgiving for both types of operations. Conversely, scheduled flights experience greater delays during the Super Bowl that have no significant difference from non-scheduled flight even with all the supporting resources. Therefore, increase in delays can likely be attributed to the increase in non-scheduled flights operating in greater numbers during the Super Bowl.

Recommendations

The results of this study provide an awareness to the effects of non-scheduled flights on scheduled flights during special events. The delays that are a result of an increase in non-scheduled flights can add millions of dollars to operating costs for air carriers and non scheduled aircraft operations. This study provides insight into an avenue for further research in mitigating delays during special circumstances in the NAS.

Further research should be conducted to determine which phases of flight cause bottlenecks and delays to occur during these special events. Studies specific to airspace, airport, and specific sectors of air traffic control may provide greater detail on how non-scheduled flights affect delays. Other special events, such as the NBAA convention, the

Daytona 500, etc., can also be studied to determine if the effects of non-scheduled flights are similar. The knowledge gained from these studies may aid government organizations and air carriers to implement changes that reduce delays, and in turn, reduce the cost of flying. Government organizations, such as the FAA, can implement new technologies that increase airspace capacity in the terminal area of airports. Air carriers or charter companies could use this study to optimize flight schedules to ensure aircraft do not arrive at the airport at the same time and cause delays.

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